

Design Decisions and Trade-offs

Project 3: Governed Deployment Pattern (cross-cloud). Azure and AWS.

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Each entry records a decision made while designing the pattern, the reasoning behind it, and the alternative that was considered and rejected.

1. Scope: a reference pattern, not a rebuild

Decision: Express Project 3 as a cloud-agnostic governance pattern with a control-to-Act mapping and concrete named services on both clouds, plus one hands-on build proof, rather than re-engineering every control on both platforms.

Rationale: A Solutions Architect is judged on architecture and documented judgment, which is where this project's weight belongs. A rigorous, transferable pattern is worth more than a partial re-implementation, and it keeps the capstone finishable and defensible.

Rejected: Fully rebuilding all ten controls on both Azure and AWS. It would multiply effort for little added signal and risk leaving the set unfinished.

2. Cross-cloud parity by design

Decision: Specify every control on both Azure and AWS, side by side.

Rationale: The pattern's value is that it is cloud-agnostic. Showing the same control on both platforms demonstrates portable architectural judgment and serves both enterprise and regulated-industry audiences.

Rejected: A single-cloud governance pattern, which would have been simpler but narrower and would not have shown platform-independent reasoning.

3. Anchor to the current, post-Omnibus timeline

Decision: Map the pattern to the Digital Omnibus-revised EU AI Act timeline: Article 50 transparency live from August 2026, high-risk obligations deferred to December 2027.

Rationale: Currency is credibility. Much published material still says August 2026 is full high-risk enforcement, which the Omnibus changed. Being precise on this signals the analysis tracks the live state of the law.

Rejected: The pre-Omnibus framing. It is now inaccurate and would undercut the whole document if an informed interviewer caught it.

4. Build to high-risk-grade controls, though the demo systems are limited-risk

Decision: Engineer the pattern to the high-risk obligations (Articles 9 to 15), even though the RAG advisor and screener classify as limited-risk.

Rationale: A pattern built to the stricter bar transfers directly to a high-risk deployment, which is where the hard requirements and the hiring demand sit. It also lets the write-up demonstrate Article 6 classification reasoning.

Rejected: Building only to the limited-risk transparency obligations the demo systems actually trigger. It would be less transferable and a weaker signal.

5. Human oversight in the critical path

Decision: Model the human-in-the-loop gate as a mandatory step before a consequential action is taken, not an optional after-the-fact review.

Rationale: Article 14 human oversight is the control that most defines a high-risk-ready system. Placing it in the critical path, before action, is what makes the oversight real rather than nominal.

Rejected: An advisory or post-hoc human review that does not gate the action, which would satisfy the letter of oversight while missing its intent.

6. One hands-on build proof: audit logging

Decision: Instantiate audit logging concretely on the existing systems as the tangible build proof for the new governance layer, rather than documenting every new control on paper only.

Rationale: Article 12 record-keeping is high-value and low-effort to demonstrate for real, and it gives genuine build evidence for a layer that Projects 1 and 2 did not have. It keeps the honest built-versus-documented line intact.

Rejected: Leaving all Project 3 controls as documentation only, which would weaken the claim that the pattern was actually exercised.

7. Model routing kept, but flagged as operational, not regulatory

Decision: Include model routing and cost control in the pattern but label it as operational governance with no direct Act obligation.

Rationale: Routing is genuinely useful enterprise governance and worth showing, but forcing it onto an Act article would be an overclaim. Flagging it honestly is the more credible move.

Rejected: Mapping routing to an Act article to make the matrix look fully covered. It would misrepresent where the law applies.

8. Honest classification and qualified rows

Decision: State plainly that the demo systems are limited-risk, and qualify the two rows that are not squarely AI Act-mandated (residency is GDPR-driven, routing is operational).

Rationale: Knowing where the law actually bites, and saying so, is a stronger credibility signal than presenting every control as Act-mandated. It is the same integrity discipline that runs through the whole portfolio.

Rejected: Presenting all controls as EU AI Act requirements, which would be neater but inaccurate.

Note on scope

This note explains the reasoning behind the pattern; the control-to-Act mapping is the artifact it produced, and the two are designed to be read together.